ATU VIRTUAL LABORATORY

P2 Full-Parallel Hybrid: learning and experiment guide

P2 hardware, command signals, SOC feedback and reproducible fuel-inventory experiments

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A practical guide from first use to test design. The system is a browser-based educational simulation; physical P2 drivetrain and vehicle validation are separate activities.

Start with the input panel

All reference presets are now inside Your experiment. Medium restores the complete 140 Nm, 2500 rpm, 70% initial SOC, 30 s reference. Run simulation calculates it, then Play response or Recorded time reveals calculated states.

Basic inputs now include torque, imposed speed, initial SOC, duration and calculation interval. The Fuel tank section adds capacity, loaded volume, density, fuel energy, price and an optional assumed-efficiency estimate. Advanced parameters remain editable with their original defaults.

Change one input at a time. Editing clears old results; Save session locally keeps the full input/calibration/fuel snapshot for this account on this browser. Reset to defaults restores the reference; it does not delete a saved lecturer record.

Understand step time without jargon

Integration time step means how far virtual time advances in one maths update. At 0.01 s there are 100 updates each simulated second; a 30 s reference contains 3001 samples including time zero.

A demand step time would be WHEN an input suddenly changes. This Phase 1 module has no demand-step-time input: requested torque and motor speed stay constant through each run. Do not interpret the integration interval as a delay before demand changes.

Playback shows existing samples; it never recalculates them or changes the integration interval. Scrubbing chooses the nearest recorded state. The final interval is shortened when necessary to end at the selected duration.

SOC protection is applied at a recorded state and affects the following integration interval. Compare 0.01 s with 0.005 s near a threshold event to study output-step resolution.

Follow the P2 hardware and signals

P2 means the electric motor joins the shared input shaft after the engine disconnect clutch and before the gearbox. An open clutch disconnects engine torque; a closed clutch allows engine and motor contributions to combine.

The interactive 3D bench contains representative engine, clutch, motor, gearbox, final drive, battery and inverter geometry, with real engine and power-electronics reference photographs. Those images and shapes do not establish manufacturer CAD or a calibrated assembly.

Drag to orbit and use camera close-ups. Shaft angle is imposed motor angular speed times selected time. Wheel/axle rotation is divided by the configured gear and final-drive ratios. Rotation is a kinematic illustration, not a calculated vehicle speed or distance. Reduced motion holds rotations still.

The signal cards show demand and SOC entering the supervisor, motor share and limits, actual engine/motor outputs and battery electrical response. Updated SOC returns to the next supervisor calculation and can disable motor traction below its threshold.

Wheel-torque error is displayed as a diagnostic comparison only. It does not return to a PID controller. Motor speed is imposed rather than controlled, and there is no physical CAN bus, sensor hardware or actuator dynamics in this module.

Fuel inventory and the optional estimate

The original Lawrence P2 equations calculate torque and battery demand, but contain no fuel map or engine fuel-flow equation. The optional fuel estimate therefore starts OFF. Fuel capacity and loaded volume are recorded even when consumption is not estimated.

Enable Estimate fuel use to study an explicitly assumed engine efficiency. Connected engine shaft kW = engine torque x clutch state x imposed angular speed / 1000. Fuel g/s = shaft kW / (assumed engine efficiency x LHV in MJ/kg). Forward interval integration accumulates mass.

Estimated litres = grams / (1000 x density in kg/L). Remaining litres = max(loaded - estimated consumed, 0). Estimated cost = consumed litres x price. Changing capacity changes available storage, not fuel economy. Changing density changes litres for the same mass.

This estimate includes only positive connected shaft output. Idle fuel, disconnected-engine operation, startup enrichment, speed/load fuel maps and thermal behavior are excluded. At zero shaft power the estimate is zero, not a validated real-engine idle claim.

An empty estimated tank produces a budget warning. It does not stop the original engine, change torque allocation or introduce battery fallback. Use the separate fuel CSV or full JSON for assumptions and samples; HybridResults CSV retains its original 13-column contract.

Read and verify the graphs

The seven original groups and 23 original channels remain. Legends use plain-language i explanations; line colour and pattern distinguish series. Move Recorded time or hover/tap a graph to inspect a sample. Zoom focuses all charts around the same selected instant.

Torque Split shows demanded, engine and motor torque. Mode and State shows motor share, clutch state, regime and SOC. Wheel Torque Tracking compares requested and delivered torque and their difference.

Motor and Power separates motor torque, imposed speed, mechanical output and battery terminal input. Battery Electrical separates power, current, terminal voltage and internal heating. Battery Energy and SOC accumulates terminal energy, loss, SOC change and RMS current. Error metrics show instantaneous and running torque mismatch.

Initial accumulated energies, RMS metrics and battery energy efficiency are zero by the reference equations. Read later samples when comparing running metrics. A mathematically defined result with a protected battery discriminant can still be physically infeasible; read warnings.

Worked tests from simple to advanced

Medium baseline: 140 Nm demand, 2500 rpm, initial SOC 0.70, duration 30 s, interval 0.01 s. Expect engine 70 Nm, motor 70 Nm, wheel torque 496.44 Nm, battery current about 61.828 A and terminal energy about

178.899 Wh.

Compare the three input-panel presets at the same speed, SOC, timing and calibration. Low / Medium / High battery terminal energies are about 153.937 / 178.899 / 303.708 Wh. Original motor and engine limits can leave positive demand unmet.

SOC test: choose Low 60 Nm and initial SOC 0.25. Motor torque becomes zero, the clutch closes and engine torque is 60 Nm. Auxiliary electrical demand remains. At initial SOC exactly 0.30, the original less-than rule permits motor traction at the first sample; discharge can trigger protection next.

Limit test: choose 400 Nm and 6000 rpm. Motor peak power reduces available motor torque and the engine can saturate at its maximum. Read wheel error and RMSE instead of assuming all demand is delivered.

Optional fuel estimate test: Medium at assumed engine efficiency 0.30, LHV 43 MJ/kg and density 0.74 kg/L gives connected shaft power about 18.326 kW and fuel flow about 1.421 g/s. In 30 s the estimate is about 0.057593 L. The original 178.899 Wh battery result stays the same.

Fuel budget test: load 0.01 L and enable the Medium estimate. The estimate exceeds the loaded amount and the remaining display reaches zero; original torque and battery calculations continue. Halving assumed engine efficiency doubles estimated fuel, while torque/battery channels remain unchanged.

Research equations, assumptions and evidence

The supervisor chooses the low motor share below the low threshold and the blended share otherwise; high and medium regimes use the same blended share in this Phase 1 formulation. Motor torque is capped by its torque and power limits, and is zero below the SOC protection threshold. The engine receives residual demand up to its torque cap.

Requested and actual wheel torque use gearbox ratio x final-drive ratio x driveline efficiency. Battery terminal power is motor shaft power divided by motor and inverter efficiencies plus auxiliary power. Current follows the protected quadratic battery equation. SOC and accumulated metrics use forward interval integration.

The original 23-channel summary and 13-column compatibility CSV remain reproducible. Full JSON also includes the separate optional fuel method, assumptions, summary and aligned sample times. Comparison uses the same core for every case.

Save your research question, inputs, fuel assumptions, advanced parameters, warning state and exports. Run the same configuration again to check reproducibility. Written equations and reference cases verify the port; source SLX and independent vehicle measurements were not supplied.

This is propulsion discharge with imposed constant speed, not a drive-cycle, regenerative, thermal, battery-ageing, engine-start, fuel-map or vehicle-distance model. Photographs, extra view controls and the optional estimate do not add those omitted dynamics.

Complete editable-input glossary

Requested torque (Nm) | Default 140. Turning effort requested before the gearbox. 140 Nm asks the engine and motor together for 140 newton metres. This demand stays constant during the run.

Motor speed (rpm) | Default 2500. How fast the motor/shared input shaft is assumed to turn. 2500 rpm means 2500 revolutions each minute. Speed is imposed and stays constant; it is not a speed-controller result.

Initial SOC (fraction) | Default 0.7. How full the battery is at the start, written as a fraction. 0.70 means 70%; 0.25 means 25%. SOC then changes as calculated current draws charge.

Simulation duration (s) | Default 30. How many virtual seconds the experiment lasts. Use the same duration when comparing energy totals. This is different from the time your computer needs to calculate it.

Integration time step (s) | Default 0.01. How much simulated time one maths update covers. 0.01 s means 100 updates per virtual second. A smaller value resolves the SOC protection boundary more closely; it is not a waiting time before demand changes.

Low torque threshold (Nm) | Default 80. Boundary between the low-demand and blended regimes. Below it, the low-torque motor share is used and the engine clutch can open.

High torque threshold (Nm) | Default 200. Boundary between medium and high research regimes. Both use the blended motor-share setting in this Phase 1 model. Must exceed the low threshold.

Low torque motor share (fraction) | Default 1. Requested fraction of input torque assigned to the motor in the low regime. 1 means 100%. Component limits still apply; the engine clutch remains open when SOC is acceptable, so a changed share can leave unmet torque.

Blended motor share (fraction) | Default 0.5. Requested motor fraction in medium and high regimes. 0.5 means 50% motor assistance before limits. The engine supplies the residual demand up to its own limit.

Engine maximum torque (Nm) | Default 142. Greatest engine torque allowed by this model. If residual demand exceeds it, delivered wheel torque can fall short.

Motor maximum torque (Nm) | Default 165. Greatest allowed motor torque. A separate peak-power limit can reduce it further at high shaft speed.

Motor peak power (W) | Default 35000. Maximum motor mechanical power. With higher imposed speed, the same power allows less torque. 35000 W is 35 kW.

SOC protection threshold (fraction) | Default 0.3. Battery fullness below which motor traction is disabled and the engine clutch closes. 0.30 means 30%. The original rule uses less than, not less than or equal to.

Gear ratio (ratio) | Default 1. Gearbox ratio between the P2 input and its output. A higher value multiplies output torque and reduces output rotation speed. No shift schedule is modeled.

Final drive ratio (ratio) | Default 3.94. Additional ratio between gearbox and wheel axle. It multiplies torque and divides rotation speed after the gearbox.

Driveline efficiency (fraction) | Default 0.9. Fraction of input torque conversion retained through the modeled driveline. 0.90 means 90%; it does not change the kinematic speed ratio.

Motor efficiency (fraction) | Default 0.92. Mechanical motor power divided by electrical motor input. 0.92 means 92%. Lower efficiency needs more battery power for the same motor shaft output.

Inverter efficiency (fraction) | Default 0.95. Fraction of inverter input power available to the motor. 0.95 means 95%. Switching waveforms are outside this model.

Auxiliary electrical load (W) | Default 500. Electrical load for other equipment in addition to motor traction. It remains in the supplied battery equation even when motor assistance is disabled.

Battery open circuit voltage (V) | Default 350. Battery voltage assumed before internal-resistance drop. It is a constant source-model parameter, not measured live hardware.

Battery internal resistance (ohm) | Default 0.045. Resistance inside the simplified battery model. It creates voltage drop and I^2R heating loss; no temperature rise is calculated.

Battery nominal energy (Wh) | Default 12000. Rated stored battery energy used to calculate charge capacity. 12000 Wh is 12 kWh. Increasing it slows SOC decline at the same current.

Angular speed epsilon (rad/s) | Default 0.001. Small positive speed safeguard in the power-to-torque limit. It avoids division by zero at rest; it is not an extra physical speed.

Numerical denominator epsilon (ratio) | Default 1e-06. Tiny positive denominator safeguard for initial and near-zero metrics. It prevents undefined ratios and does not represent a physical power loss.

Fuel tank capacity (L) | Default 55. Maximum fuel volume the tank can hold. A 55 L tank can start partly filled. Tank size does not improve engine efficiency or change the torque/battery equations.

Fuel currently in tank (L) | Default 20. Litres actually loaded at the start. Must not exceed tank capacity. Remaining fuel can only be estimated when the optional fuel estimate is enabled.

Fuel density (kg/L) | Default 0.74. Mass of one litre of fuel. At 0.74 kg/L, a litre weighs 740 g. This converts estimated consumed mass into litres.

Fuel lower heating value (MJ/kg) | Default 43. Chemical energy stored in each kilogram of fuel. Use a value appropriate to the fuel. It is an explicit assumption for the optional estimate, not part of the original P2 equations.

Assumed engine efficiency (fraction) | Default 0.3. Assumed fraction of fuel chemical power reaching the connected engine shaft. 0.30 means 30%. This constant estimate has no engine map, idle fuel or startup enrichment.

Fuel price per litre (currency/L) | Default 0. Your chosen price for one litre. Estimated cost equals estimated consumed litres times this price. Zero means the run has not been priced.

Analytical signal definitions

Time (s). Recorded virtual time, in seconds.

Requested torque (Nm). Turning effort requested before the gearbox. 140 Nm asks the engine and motor together for 140 newton metres. This demand stays constant during the run.

Engine torque (Nm). Allocated engine torque after its limit. Actual contribution reaches the shared shaft only with the clutch closed.

Motor torque (Nm). Actual motor torque after share command, motor torque/power limits and SOC protection.

Motor share command (fraction). Commanded motor fraction before component limits. 0.5 asks for half the requested torque; actual torque can be lower.

Clutch state (code). 1 means closed and mechanically coupled; 0 means open and disconnected. P2 places this clutch before the motor and transmission.

Torque regime (code). Research classification: 1 low, 2 medium, 3 high demand. This is not a vehicle diagnostic trouble code.

Requested wheel torque (Nm). Requested input torque multiplied by gear ratio, final-drive ratio and driveline efficiency.

Actual wheel torque (Nm). Connected engine torque plus motor torque, multiplied by both ratios and driveline efficiency.

Wheel torque error (Nm). Requested minus delivered wheel torque. This is a diagnostic monitor; the model does not use it to command a PID correction.

Motor angular speed (rad/s). The imposed motor speed converted from rpm into radians per second. It is held constant.

Motor mechanical power (W). Actual motor torque times imposed shaft angular speed. This is mechanical output power, in watts.

Battery terminal power (W). Motor electrical demand plus auxiliary load at the battery terminals. Internal battery loss is tracked separately.

Battery current (A). Calculated discharge current from terminal-power demand, source voltage and internal resistance. Protected discriminants are warned about.

Battery terminal voltage (V). Source voltage minus current times internal resistance. It is a calculated terminal voltage, not a measured sensor.

Internal loss power (W). Current squared times internal resistance. This is internal heating power; temperature is not simulated.

Battery SOC (%). Calculated remaining battery charge as a percentage. This state feeds the next supervisor decision.

Delta SOC (percentage points). Initial minus current SOC, in percentage points. From 70% to 68% is a two-point reduction.

Battery energy used (Wh). Accumulated terminal electrical energy in watt-hours. Internal loss is separate.

Battery loss energy (Wh). Accumulated internal battery heating energy in watt-hours.

Battery energy efficiency (%). Terminal energy divided by terminal plus internal-loss energy, times 100. This is not complete fuel-to-wheel efficiency.

RMS battery current (A). Root-mean-square current over elapsed simulation time; a useful load/heating metric.

Torque RMSE (Nm). Root-mean-square wheel-torque error over time. Zero means exact tracking under the written model, not physical vehicle validation.

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